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***fiber_context* - fibers without scheduler**

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abstract

This paper proposes a minimal API that enables stackful context switching **without** the need for a **scheduler**. The API is suitable to act as building-block for high-level constructs such as stackful coroutines as well as cooperative multitasking (aka user-land/green threads that incorporate a **scheduling facility**).

This revision addresses concerns, questions and suggestions from the past meetings. The proposed API supersedes the former proposals N3985,⁸ P0099R1,¹¹ P0534R3¹² and P0876R19.³³

Because of name clashes with *coroutine* from C++20, *execution context* from executor proposals and *continuation* used in the context of `future::then()`, the committee has indicated that *fiber* is preferable. However, given the foundational, low-level nature of this proposal, we choose *fiber_context*, leaving the term *fiber* for a higher-level facility built on top of this one.

Informally within this proposal, the term *fiber* is used to denote the flow of control launched and represented by the first-class object `fiber_context`.

It's telling that when Hana Dusikova was working on implementations of P3367R3 `constexpr` coroutines,³⁸ the "easiest way to model a coroutine," the "obvious first choice," was to use fibers in the `constexpr` evaluator.

Revision History

This document supersedes P0876R19.

Changes since P0876R19

- Add information about implementability of per-fiber exception state.
- Add links to St. Louis 2024 EWG notes, Wrocław 2024 EWG notes and Microsoft implementability email.
- Add discussion of P3620R0.
- Mention P3367R3 `constexpr` coroutines.

Changes since P0876R18

- Move exception state test programs to Appendices.
- Link Boost.Context patch that produces correct fiber-specific exception behavior on Windows and Linux using `libstdc++`.
- Add references to six additional production libraries built on fiber technology.

Changes since P0876R17

- Distinguish between a *prepared* and a *suspended* fiber.
- Distinguish the two context switches implied by entry to, and return from, `resume_with()`.
- Remove `current_exception_within_fiber()`, which became moot in P0876R17.

Changes since P0876R16

- Update to reference N4981.
- Add `<fiber_context>` header file to headers table.
- Remove `resume_with()` "Case A" and "Case B" in favor of nested bullet lists. Fix a bug in definition of `internal_resume`.
- Revert `resume_with()` *Returns:* and *Throws:* clauses to R15 structure, eliminating "Case C" and "Case D".
- Use scoped exposition-only terms *calling fiber*, *target fiber* and *previous fiber* instead of quoting the phrases. Give previous fiber definition its own bullet.
- Eliminate *internal_resume* parameter *after*, also definitions of `after_entry_copy`, `after_stack_copy` and `after_deleter_copy`. Describe *internal_resume* in terms of the currently running fiber.
- Explicitly state that *internal_resume* is exposition only, and italicize references.
- Move predicate for first *internal_resume* definition to start of bullet text. Don't state the inverse predicate for the second.
- Remove one level of bullet list nesting from the second *internal_resume* definition. Sequence the bullet list by appending ", then" after each item.
- Per EWG in St. Louis, remove implementation defined meaning of *currently handled exception* and `uncaught_exceptions`. Now both are fiber-specific.
- Clarify explicit constructor *Throws:* clause.

Changes since P0876R15

- [fiber.context.overview] is now “Overview” instead of “Preamble”.
- Make default `fiber_context` constructor = `default`. Remove its section from member descriptions.
- In unary constructor, move “Mandates” before “Constraints”. The “Preconditions” entry is actually a Constraint: move it and remove “Preconditions”.
- In span constructor, stated “Preconditions” are actually “Constraints”.
- In constructor descriptions, use less precise language about copying `entry`, `stack` and `deleter`. Add Note about them not being `fiber_context` members. Move mention of `stack` to a Note.
- Rephrase `resume_with()` Note about emptying its `fiber_context` object to avoid the appearance of a normative statement.
- Remove mention of “legacy behaviour” from `current_exception_within_fiber()`.
- Remove mention of “thread of execution” from “abstract,” “control transfer mechanism” and the section on `std::uncaught_exceptions()` and `std::current_exception()`.
- Simplify definitions of implicit fiber vs. explicit fiber.
- Add [intro.fibers] statement that a thread is always running one fiber, but can switch between fibers. This replaces the more detailed description of what happens when a fiber calls `resume()` or `resume_with()`.
- In [intro.fibers], hoist “owning thread” definition to its own paragraph 3 and clarify.
- Remove assertion that a fiber is an execution agent.
- Modify [except.throw] paragraphs 2 and 4, and [except.handle] paragraph 6, to constrain exception propagation to a fiber.
- Describe explicit fiber as being “prepared,” with a statement that it comes into existence on first resumption.
- Remove a few stray instances of “may”.
- Move assertion that a received `fiber_context` object could represent either an explicit fiber or an implicit fiber to a Note.
- Move assertion that no `fiber_context` object represents a running fiber up to Overview.
- Use `successor` rather than the more generic `continuation` to reference the `fiber_context` object returned by a terminating fiber.
- Remove nesting from `resume_with()` *Throws*..
- Remove Note that the caller of `resume_with()` can detect whether the previous fiber has terminated: not necessarily.
- Hoist section on `std::uncaught_exceptions()` and `std::current_exception()` to have its own table of contents entry. Extend with examples of bad behavior when switching out of a catch block to a fiber which itself catches some exception before switching back to the original fiber.
- Remove explicit `delete` declarations of copy constructor and copy assignment: these are implicitly deleted.
- Since we want the constructor’s `entry` and `deleter` parameters to support move-only objects, remove *Cpp17CopyConstructible* requirements.
- For the same reason, state that `entry_copy` and `deleter_copy` are initialized rather than copied.
- Therefore “any exception from initialization of `entry_copy`” and the same for `deleter_copy`.
- Remove mention of “function call stack” from constructor *Throws*..
- `stack.data()` and `stack.size()` must meet implementation requirements, not the `span<byte>` itself.
- Remove *Postconditions*:`other.empty()` from move constructor and move assignment: these are implied by definition.
- Move statement about UB from stack overflow to `fiber_context` Overview.

- Modify example about early destruction of exceptions to add sequence comments, highlight access to a destroyed exception object.
- Fix erroneous [fibercontext.mumble] references in class comments.
- Add green changebars for entirely new sections.
- Remove `std::` qualification from `decay_t` in *Effects:*.
- Remove the destructor Note about encouraging a fiber to terminate voluntarily.
- Clarify that `current_exception_within_fiber()` is `true` if `std::current_exception()` reports exceptions “only” within the current fiber. Remove `constexpr`: compiler can produce object code that might be linked with alternative runtimes.
- Remove “.” after “;”.
- `resume_with()` *Mandates*: `is_invocable_r<...>` is `true`. Add periods to *Mandates:* and *Preconditions:*.
- Add *Preconditions:* to span constructor that deleter must not throw. Remove cleanup exceptions from `resume_with()` *Throws:*. Remove “before this point, no exceptions” bullet in *Effects:*.
- `resume_with()` evaluates `invoke_r(fn)`. Merge Notes about what its returned can be.
- Substantially rework `resume_with()` description. Break out and label the four cases: (target not yet entered, target previously suspended); (previous exited, previous called `resume_with()`). Use case labels in *Effects:*, *Returns:* and *Throws:*. Break out internal-resume operation because it’s self-referential.
- Add span constructor *Preconditions:* for `decay_t<D>` meeting *Cpp17MoveConstructible* requirements.
- Remove `can_resume()` Note about “can resume.”
- For `resume()`, use `std::identity` instead of identity lambda.

Changes since D0876R15

- Updated to reference N4971.
- Inserted a section to clarify relationship between threads and fibers.
- Borrowed “single flow of control” definition for “fiber.”
- Added Note clarifying “flow of control” as state, with reference to [\[stacktrace.general\]](#).
- Changed stacktrace “invocation sequence” to reference “fiber” rather than “thread of execution.”
- Changed “thread” definition to be the execution agent that runs fibers.
- Clarified that if a fiber terminates by returning an empty `fiber_context` instance, `std::terminate` is called.
- Added `constexpr fiber_context::current_exception_within_fiber`.
- Removed definition of “function call stack.”
- Removed change to definition of expression evaluation conflict.
- Removed Note about the second fiber in the program.
- Changed “`fiber_context` instance” to “`fiber_context` object.”
- Changed “method” to “member function.”
- Removed paragraph numbers from internal cross-references.
- Clarified editorial directives amongst not-green new text.
- Used “fiber.context” in stable labels.
- Changed the lone remaining preamble section in [fiber.context] from “Empty vs. Non-Empty” to “Preamble.”
- Moved to “Preamble” the 1:1 relationship between non-empty `fiber_context` objects and suspended fibers.
- Used “Effects: Equivalent to `return <expression>`” for `empty()` and `operator bool()`.
- Referenced `main` instead of `main()`.

Changes since P0876R14

- Invoked “blocks with forward progress guarantee delegation” words of power for `resume_with()`, guaranteeing mutual exclusion.
- Fixed Mandates and Throws concerning the entry function and deleter passed to the implicit-stack or explicit-stack constructor.
- Cleaned up wording around initializing, assigning and testing the exposition-only `state` member.
- Dampened the optimism of the proposed feature-test macro.

Changes since P0876R13

- At LEWG’s request, retracted changes to `uncaught_exceptions()` and `current_exception()`, instead clarifying that results may reflect exceptions on other fibers running on the current thread.
- Updated against draft standard N4958.
- Deleted “User-Mode” from new section title “Cooperative Threads” and removed the explanatory paragraph.
- Removed `explicit` from the explicit-stack constructor.
- Added `system_error: resource_unavailable_try_again` to the *Throws:* clause of the implicit-stack constructor.
- Changed `bad_alloc` to `system_error: resource_unavailable_try_again` in the *Throws:* clause of the explicit-stack constructor.
- Stated that the move constructor and move assignment operator empty the moved-from `fiber_context`.
- Removed the `empty()` precondition from assignment operator; instead added the same `(! empty())` effect as for the destructor.
- Removed `resume_with()` references to “execution context.” Existing section 7.6.1.3 Function call `[expr.call]` makes no mention of saving or restoring state.
- Removed bullets in `resume_with()` *Returns:* and *Throws:* clauses regarding `resume()`, since they can be inferred from `resume_with()` and the trivial-lambda equivalence described for `resume()`.
- Removed the *Remarks:* about concurrent calls from multiple threads from `can_resume()`, leaving in place the editorial note about the intentional absence of `const`.
- Changed exposition-only `state` member from unspecified-type to `void*`.
- Sanitized stable names.
- Moved feature-test macro to appropriate section.
- Cleaned up the header-file synopsis.
- Grouped class members with forward references.
- Added `std::swap()` specialization.
- Added obtrusive paragraph numbers.
- Streamlined single-item dash lists.
- Changed *Ensures* to *Postconditions*.
- Changed template parameters from `typename` to `class`.
- Tweaked constructor *Preconditions:* / *Mandates:*.
- Clarified that `entry_copy`, `stack_copy` and `deleter_copy` are not intended to be data members of `fiber_context`.
- Streamlined initialization of these exposition objects.
- “Instantiates a `fiber_context`” => “Initializes state”
- `empty()` returns `true` => `empty()` is `true`, et al.
- Removed explicit-stack constructor *Preconditions:* for stack size and alignment, since *Throws:* explicitly specifies exceptions for violations.

- Rephrased *Effects*: of move constructor.
- Extracted “Let” statements from *Effects*: to preceding paragraphs.

Changes since P0876R12

- Proposed that `uncaught_exceptions()` and `current_exception()` be specific to the current thread of execution.
- Specified that constructors *decay-copy* the entry-function.
- Changed `span<byte, N>` constructor param to simply `span<byte>`; also accepted deleter function, which it must *decay-copy*.
- Specified constructor exceptions.
- Specified that destroying a non-empty `fiber_context` calls `terminate()`.
- Clarified that when `resume_with()` is called, `empty()` becomes true immediately.
- Introduced exposition-only `fiber_context::state` member to streamline wording.
- Removed `concurrency_v2` namespace.
- Changed “Equivalent to” to “As-if”.
- Clarified Preconditions vs. Mandates.

Changes since P0876R11

- Removed `get_stop_source()`, `get_stop_token()`, `request_stop()` and exposition-only `ssource` members.
- Added a `fiber_context` constructor accepting a caller-provided uninitialized memory area for the new fiber’s function call stack.

Bundling a `stop_source` into `fiber_context` presented implementability concerns. Although each fiber (specifically, its function call stack) is itself a persistent entity, the `fiber_context` representing that fiber is not: a new `fiber_context` object is synthesized on every suspension. This presents a problem: how does the code that suspends a fiber find its associated `stop_source` shared state?

A consumer wishing to pass a `std::stop_token` to a new fiber can itself instantiate `std::stop_source`, obtain from it a `stop_token` and bind that `stop_token` in a lambda passed to the `fiber_context` constructor. Accordingly, the `fiber_context` API need not explicitly support that.

Changes since P0876R10

- Removed `cancel()` method and the cancellation-function constructor argument. Replaced with the `std::jthread` stop token handling API: `get_stop_source()`, `get_stop_token()` and `request_stop()`. This simplifies examples by eliminating `launch()` and `assert_on_cancel`.
- Added a section exploring the relationship of `fiber_context` to the larger C++ ecosystem.
- Reordered some sections to make the paper more accessible for new readers.

Changes since P0876R9

- Removed `resume_from_any_thread()`, `resume_from_any_thread_with()`, `cancel_from_any_thread()` and `can_resume_from_this_thread()`, along with stated support for resuming a suspended fiber on some thread other than the one on which it was launched.

In Belfast, EWG came down strongly against cross-thread fiber resumption. The most emphatic objection was that for a function referencing TLS, multiple compilers cache TLS pointers on the function’s stack frame. Resuming a fiber containing that stack frame on some other thread would cause problems. In the best case, the resumed function would merely reference TLS belonging to the wrong thread – but at some point the original thread will terminate, its TLS will be destroyed, and the cached pointers will be left dangling.

With `fiber_context`, any opaque function call might possibly suspend – but invalidating cached TLS pointers across every opaque function call is deemed unacceptable overhead.

Changes since P0876R8

- Reinstated cancellation function constructor argument.
- Added `cancel()` and `cancel_from_any_thread()` member functions.
- Re-removed `std::unwind_fiber()`.

SG1 directed P0876R9 to conform to the Cologne 2019 recommendations, with any other changes proposed in a separate paper.

Changes since D0876R7

- Cancellation function removed from `fiber_context` constructor.
- `std::unwind_fiber()` re-added, with implementation-defined behaviour.
- Added elaboration of `filament` example to bind cancellation function.

P0876R8 diverged from the recommendations of the second SG1 round in Cologne 2019. It did not introduce `cancel()` or `cancel_from_any_thread()` member functions. In fact it removed the cancellation-function constructor argument.

`fiber_context` is intended as the lowest-level stackful context-switching API. Binding a cancellation-function on the fiber stack is a flourish rather than a necessity. It adds overhead in both space (on the fiber stack) and time (to traverse the stack to retrieve the cancellation-function). For this API, it should suffice to pass the desired cancellation-function to `resume_with()`. If it is important to associate a cancellation-function with a particular fiber earlier in the lifespan of the fiber, a struct serves.

A more compelling reason to avoid constructing an explicit fiber with a cancellation-function is that no implicit fiber has any such cancellation-function – and the consuming application cannot tell, a priori, whether a given `fiber_context` instance represents an explicit or an implicit fiber. If `*this` represents an implicit fiber, what should the proposed `cancel()` member function do?

Passing a specific cancellation-function to `resume_with()` avoids that problem.

P0876R8 follows SG1 recommendation in making it Undefined Behaviour to destroy (or assign to) a non-empty `fiber_context` instance.

`std::unwind_fiber()` was reintroduced with implementation-defined behaviour to allow fiber cleanup leveraging implementation internals. Its use was entirely optional (and auditable).

Changes since P0876R6

- Implicit stack unwinding (by non-C++ exception) removed.
- `std::unwind_fiber()` removed.
- Cancellation function added to `fiber_context` constructor.

In Cologne 2019, SG1 took the position that:

- The `fiber_context` facility is not the only C++ feature that requires “special” unwinding (special function exit path).
- Such functionality should be decoupled from `fiber_context`. It requires its own proposal that follows its own course through WG21 process.
- Depending on this (yet to be written) proposal would unduly delay the `fiber_context` facility.
- For now, the `fiber_context` facility should adopt a “less is more” approach, removing promises about implicit unwinding, placing the burden on the consumer of the facility instead.
- This leaves the way open for `fiber_context` to integrate with a new, improved unwind facility when such becomes available.

The idea of making `fiber_context`'s constructor accept a cancellation function was suggested to permit consumer opt-in to P0876R5 functionality where permissible, or convey to the fiber in question by any suitable means the need to clean up and terminate.

Requiring the cancellation function is partly because it remains unclear what the default should be. This could be one of the questions to be answered by a TS. Moreover, the absence of a default permits specifying later that the default engages the new, improved unwind facility.

Changes since P0876R5

- `std::unwind_exception` removed.
- `fiber_context::can_resume_from_any_thread()` renamed to `can_resume_from_this_thread()`.
- `fiber_context::valid()` renamed to `empty()` with inverted sense.
- Material has been added concerning the top-level wrapper logic governing each fiber.

`std::unwind_exception` was removed in response to deep discussions in Kona 2019 of the surprisingly numerous problems surfaced by using an ordinary C++ exception for that purpose.

Problems resolved by discarding `std::unwind_exception`:

- When unwinding a fiber stack, it is essential to know the subsequent fiber to resume. `std::unwind_exception` therefore bound a `fiber_context`. `fiber_context` is move-only. But C++ exceptions must be copyable.
- It was possible to catch and discard `std::unwind_exception`, with problematic consequences for its bound `fiber_context`.
- Similarly, it was possible to catch `std::unwind_exception` but not rethrow it.
- If we attempted to address the problem above by introducing a `std::unwind_exception` operation to extract the bound `fiber_context`, it became possible to rethrow the exception with an empty (moved-from) `fiber_context` instance.
- Throwing a C++ exception during C++ exception unwinding terminates the program. It was possible for an exception implementation based on `thread_local` to become confused by exceptions on different fibers on the same thread.
- It was possible to capture `std::unwind_exception` with `std::exception_ptr` and migrate it to a different fiber – or a different thread.

P3620R0: Concerns with the proposed addition of fibers to C++26

At a high level, P3620R0³⁹ appears to argue that unless fibers are appropriate for all use cases, they must not be available for any use case. This ignores the industry experience cited in [fiber_context as building block for higher-level frameworks](#).

Not every C++ feature is applicable to every environment. `breakpoint()` is not generally found in production code. A library that writes to `std::cerr` will cause problems for an application running in a windowed environment that has no `stderr` file handle. A library that throws exceptions is a poor choice for an application that forbids exceptions. A library that creates `std::threads` will cause trouble for an application that's not expecting them.

Fibers are not lightweight threads P3620R0 states that operating system vendors have largely abandoned attempts to support fibers as N:M threading, because operating system threads have more state than it's feasible to manage with fibers.

`fiber_context` does not claim to support lightweight threads. `fiber_context` is a tool for organizing the flow of control within an operating system thread. It does not need to manage signals, signal masks or other facilities beyond the C++ abstract machine.

TLS P3620R0 notes that `thread_local` storage is shared between all the fibers on a thread. P3346R0³⁷ proposed to modify `thread_local` to mean fiber-specific. This was rejected by SG1 in Wrocław.⁶²

This semantic can nonetheless be addressed by a higher-level library. For instance, `Boost.Fiber`⁴⁴ provides `fiber_specific_ptr`.

P3620R0 further claims that C++20 coroutines do not have this problem. Actually, they do. If, on entry, a coroutine links an object into a linked list anchored with a `static` or `thread_local` pointer, then unlinks it on final return, reaching that coroutine from different interleaved invocation sequences will corrupt that linked list. This issue did not block adoption of C++20 coroutines.

It may be worth noting that coroutines provide no entity analogous to a fiber. It would not be straightforward to support chain-of-coroutines-local storage.

Deadlocks P3620R0 points out that switching fibers within a thread while holding a lock may lead to accidental deadlock.

This semantic can be addressed by a higher-level library. For instance, `Boost.Fiber`⁴⁴ provides fiber-aware synchronization primitives such as `boost::fibers::mutex`.

C++20 coroutines have the same problem. This issue did not block adoption of C++20 coroutines.

It would not be straightforward to support chain-of-coroutines-aware synchronization primitives.

`fiber_context` and the larger C++ ecosystem

higher-level libraries `fiber_context` [as building block for higher-level frameworks](#) enumerates a number of higher-level abstraction libraries built upon the *Boost.Context* implementation of the API proposed in this paper. This is not an exhaustive list, but it suffices to illustrate that there is widespread interest in this functionality.

The most significant point about this proposal is that, given `fiber_context`, all those libraries can be written in standard C++. They need not themselves be integrated into the Standard.

Because it creates and switches between different function call stacks, though, the `fiber_context` facility cannot be written in portable C++. There is real value to integrating this library into the Standard.

Boost.Context is maintained by one individual to support the specific set of processors and operating systems to which he has access. The `fiber_context` facility will ensure support in every implementation of the C++ runtime, extending into the future.

Given the lively ecosystem of open-source libraries, it's possible that standardizing `fiber_context` could suffice. It is not essential that WG21 must standardize additional higher-level libraries before the facility would become useful. The uptake of *Boost.Context* illustrates that the community can make good use of `fiber_context`.

However, the evolution of this proposal and the WG21 discussions thereof have surfaced a number of interesting adjacencies.

cancellation Given C++ support for concurrency, in various forms, within a program, cancellation of an asynchronous task remains a topic of widespread interest. It has been much discussed, e.g. in P1677R2,³⁴ P1820R0³⁵ and P2175R0.³⁶

Previous revisions of this paper have proposed canceling a suspended fiber by injecting an exception, e.g. using `fiber_context::resume_with()`. A comparable approach was rejected for `std::jthread`, although it's worth noting that cooperative fibers differ in a very significant respect: every fiber suspends at a well-defined point, namely a call to `resume_with()`.*

Evolution of the exception mechanism itself¹⁴ may affect the viability of using exceptions for cancellation.

This paper simply notes that an invoker can use lambda binding to pass (e.g.) a `std::stop_token` from the Standard,⁹ section 33.3, to a fiber at launch time.

modules and optimizations Before modules, the only information the compiler could know about a function in an external translation unit was what a human coder stated in the relevant header file. But since the information in a module is prepared by the compiler itself, a subsequent compile of a translation unit that imports that module can know as much about each module function as it would if the function's source code was found within the current translation unit.

This permits the compiler to infer and propagate attributes. If a function neither contains a throw statement nor calls other functions, the compiler can conclude that it doesn't throw. It can encode this information in the module produced for that translation unit, so that subsequent compiles can make use of the knowledge. If another function contains no throw statement and calls only functions known not to throw, it too can be implicitly marked `nothrow`.

Similarly, when compiling a function that can never return, the compiler can so indicate in the output module. Any caller whose code path leads unconditionally to any such function can also be known never to return.

In much the same way, the module describing the library's `fiber_context::resume_with()` method can mark it as *can-suspend*. Then any caller of `resume_with()` will also be marked *can-suspend*, and so forth. The compiler can use this to improve its optimization tactics around any call to a *can-suspend* function.

(The *can-suspend* characteristic of a `co_await` coroutine function is just as pervasive, but in that case the coder must manually propagate it.)

synchronization primitives The Standard⁹ provides an assortment of primitives for synchronizing work between threads, e.g. sections 33.6, 33.7, 33.8, 33.9, 33.10. An essential behaviour of many such synchronization primitives is to pause, or suspend, execution of the current thread until some external condition is satisfied.

Such suspension is very different from fiber suspension as proposed in this paper. This proposal neither requires nor implies a scheduler. A fiber suspends by explicitly designating the next fiber to resume, either by passing its `fiber_context` to `resume_with()` or by returning that `fiber_context` from its entry-function.

C++ threads, in contrast, assume a thread scheduler, usually provided by the operating system. Suspending a thread means passing control to the scheduler, which reallocates CPU resources to other pending threads. At some future time, the scheduler is responsible for directing some CPU core to resume the suspended thread.

Fiber suspension as implemented by `fiber_context` is independent of thread suspension. Suspending the running fiber simply means directing the thread to run a different fiber; the thread continues running. Conversely, suspending the host thread (e.g. by invoking a synchronization primitive) means that *no* fiber is running on that thread.

A higher-level fiber-based library that emulates the `std::thread` API, such as *Boost.Fiber*,⁴⁴ necessarily implements a fiber scheduler, permitting implicit fiber suspension. Standardizing such a library would raise the interesting question of how to present fiber-aware synchronization primitives.

A straightforward approach is to present a suite of fiber-aware synchronization primitives distinct from, but analogous to, the thread-based synchronization primitives.[†] A program running multiple fibers within a thread would use fiber-aware synchronization primitives rather than thread-based synchronization primitives. Evaluating a thread-based synchronization primitive would suspend the entire thread, as usual, halting all fibers within that thread.

It is tempting to contemplate modifying the semantics of the present suite of synchronization primitives to make them fiber-aware. Naturally this is a matter of some concern.

For purposes of this `fiber_context` proposal, though, it is entirely moot.

*Although exception-based cancellation is not implicitly supported, a consumer of `fiber_context` may still explicitly pass to `resume_with()` an invocable that raises an exception in the suspended fiber.

[†]This is the approach taken by *Boost.Fiber*.

Execution Agent Local Storage A similar question arises concerning variable storage duration. Should the Standard introduce a fiber-specific storage duration, e.g. `fiber_local`, analogous to `thread_local`⁹? (section 6.7.5.3 **Thread storage duration**)

The Standard defines the general term *execution agent* (section 33.2.5.1) to allow for multiple kinds of parallelism. It seems reasonable to assume that over time, new types of execution agents will be defined. Will we want the Standard to present a new `xyz_local` storage duration for each new “xyz” execution agent type?

P0772R1¹⁵ notes that library code should not have to care what kind of execution agent is running it. Already it’s important to ensure that library code avoids `static` variables because any such variable prohibits calling that library from more than one thread. P0772R1 suggests a generalized variable storage duration dynamically local to the innermost current execution agent.

(The same consideration about library code impacts the above question about presenting fiber-aware synchronization primitives.)

It’s true that if:

- a particular function relies on a `thread_local` variable
- the function calls a function that resumes a different fiber
- the other fiber makes a different call to that same function, or to another function that modifies the same `thread_local` variable

then on resumption of the original fiber, the function will observe the new value for the `thread_local` variable.

This is analogous to use of a `static` variable by multiple threads in the same program – though not as bad, since it doesn’t produce race-related Undefined Behaviour on top of correctness problems.

`std::thread` was introduced despite this problem because it’s *useful*.

Multiple C++ implementations cache a pointer to thread-local storage in the stack frame of a function referencing TLS. If a suspended fiber were resumed by a thread other than the one on which it previously ran, such cached TLS pointers would point to TLS for the wrong thread. This is why such cross-thread resumption is forbidden.

(This is the only optimization that has yet been surfaced by implementers as a potentially problematic interaction with fibers.)

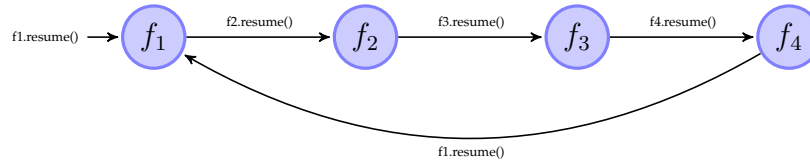
tooling One particularly valuable consequence of adding `fiber_context` to the Standard will be to add fiber awareness to debuggers, performance analyzers and other tools that inspect a running C++ program.

Such tools need only be aware of `fiber_context`. They would *not* need to be further adapted to support higher-level libraries built on the `fiber_context` facility.

control transfer mechanism

According to the literature,⁷ coroutine-like control-transfer operations can be distinguished into the concepts of *symmetric* and *asymmetric* operations.

symmetric fiber A symmetric fiber provides a single control-transfer operation. This single operation requires that the control is passed explicitly between the fibers.

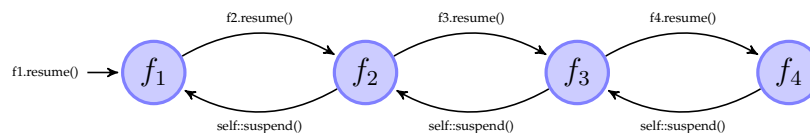


```
1 fiber_context* pf1;
2 fiber_context f4{[&pf1]{
3     pf1->resume();
4 }};
5 fiber_context f3{[&f4]{
6     f4.resume();
7 }};
8 fiber_context f2{[&f3]{
9     f3.resume();
10 }};
11 fiber_context f1{[&f2]{
12     f2.resume();
13 }};
14 pf1=&f1;
15 f1.resume();
```

In the pseudo-code example above, a chain of fibers is created.

Control is transferred to fiber f_1 at line 15 and the lambda passed to constructor of f_1 is entered. Control is transferred from fiber f_1 to f_2 at line 12 and from f_2 to f_3 (line 9) and so on. Fiber f_4 itself transfers control directly back to fiber f_1 at line 3.

asymmetric fiber Two control-transfer operations are part of asymmetric fiber's interface: one operation for resuming (`resume()`) and one for suspending (`suspend()`) the fiber. The suspending operation returns control back to the calling fiber.



```
1 // hypothetical API
2 fiber_context f4{[] {
3     self::suspend();
4 }};
5 fiber_context f3{[&f4]{
6     f4.resume();
7     self::suspend();
8 }};
9 fiber_context f2{[&f3]{
10    f3.resume();
11    self::suspend();
12 }};
13 fiber_context f1{[&f2]{
14    f2.resume();
15    self::suspend();
16 }};
```

```
17 f1.resume();
```

In the pseudo code above execution control is transferred to fiber `f1` at line 16. Fiber `f1` resumes fiber `f2` at line 13 and so on. At line 2 fiber `f4` calls its suspend operation `self::suspend()`. Fiber `f4` is suspended and `f3` resumed. Inside the lambda, `f3` returns from `f4.resume()` and calls `self::suspend()` (line 6). Fiber `f3` gets suspended while `f2` will be resumed and so on ...

The asymmetric version needs **N-1 more** fiber switches than the variant using symmetric fibers.

While asymmetric fibers establish a caller-callee relationship (strongly coupled), symmetric fibers operate as siblings (loosely coupled).

Symmetric fibers represent independent flows of control, making symmetric fibers a suitable mechanism for concurrent programming. Additionally, constructs that produce sequences of values (*generators*) are easily constructed out of two symmetric fibers (one represents the caller, the other the callee).

Asymmetric fibers incorporate additional fiber switches as shown in the pseudo code above. It is obvious that for a broad range of use cases, asymmetric fibers are less efficient than their symmetric counterparts.

Additionally, the calling fiber must be kept alive until the called fiber terminates. Otherwise the call of `suspend()` will be undefined behaviour (where to transfer execution control to?).

Symmetric fibers are more efficient, have fewer restrictions (no caller-callee relationship) and can be used to create a wider set of applications (generators, cooperative multitasking, backtracking ...).

fiber_context as a first-class object

Because the symmetric control-transfer operation requires explicitly passing control between fibers, fibers must be expressed as *first-class objects*.

Fibers exposed as first-class objects can be passed to and returned from functions, assigned to variables or stored into containers. With fibers as first-class objects, a program can **explicitly control the flow of execution** by suspending and resuming fibers, enabling control to pass into a function at exactly the point where it previously suspended.

Symmetric control-transfer operations require fibers to be first-class objects. First-class objects can be returned from functions, assigned to variables or stored into containers.

encapsulating the stack

Each fiber is associated with a function call stack and is responsible for managing the lifespan of its stack (allocation at construction, deallocation when fiber terminates). The RAII-pattern* should apply.

Copying a `fiber_context` must not be permitted!

If a `fiber_context` were copyable, then its stack with all the objects allocated on it must be copied too. That presents two implementation choices.

- One approach would be to capture sufficient metadata to permit object-by-object copying of stack contents. That would require dramatically more runtime information than is presently available – and would take considerably more overhead than a coder might expect. Naturally, any one move-only object on the stack would prohibit copying the entire stack.
- The other approach would be a bitwise copy of the memory occupied by the stack. That would force undefined behaviour if any stack objects were RAII-classes (managing a resource via RAII pattern). When the first of the fiber copies terminates (unwinds its stack), the RAII class destructors will release their managed resources. When the second copy terminates, the same destructors will try to doubly-release the same resources, leading to undefined behaviour.

*resource acquisition is initialisation

A fiber API must:

- encapsulate the stack
- manage lifespan of an explicitly-allocated stack: the stack gets deallocated when `fiber_context` goes out of scope
- prevent accidentally copying the stack

Class `fiber_context` must be *move-only*.

invalidation at resumption

The framework must prevent the resumption of an already running or terminated (computation has finished) fiber.

Resuming an already running fiber will cause overwriting and corrupting the stack frames (note, the stack is not copyable). Resuming a terminated fiber will cause undefined behaviour because the stack might already be unwound (objects allocated on the stack were destroyed or the memory used as stack was already deallocated).

As a consequence each call of `resume()` will empty the `fiber_context` instance.

Whether or not a `fiber_context` is empty can be tested with member function `operator bool()`.

To make this more explicit, functions `resume()` and `resume_with()` are rvalue-reference qualified.

The essential points:

- regardless of the number of `fiber_context` declarations, exactly one `fiber_context` instance represents each suspended fiber
- no `fiber_context` instance represents the currently-running fiber

Section [solution: avoiding non-const global variables and undefined behaviour](#) describes how an instance of `fiber_context` is synthesized from the active fiber that suspends.

A fiber API must:

- prevent accidentally resuming a running fiber
- prevent accidentally resuming a terminated fiber
- `resume()` and `resume_with()` are rvalue-reference qualified

problem: avoiding non-const global variables and undefined behaviour

According to C++ *core guidelines*,⁴⁰ non-const global variables should be avoided: they hide dependencies and make the dependencies subject to unpredictable changes.

Global variables can be changed by assigning them indirectly using a pointer or by a function call. As a consequence, the compiler can't cache the value of a global variable in a register, degrading performance (unnecessary loads and stores to global memory especially in performance critical loops).

Accessing a register is one to three orders of magnitude faster than accessing memory (depending on whether the cache line is in cache and not invalidated by another core; and depending on whether the page is in the TLB).

The order of initialisation (and thus destruction) of static global variables is not defined, introducing additional problems with static global variables.

A library designed to be used as building block by other higher-level frameworks should avoid introducing global variables. If this API were specified in terms of internal global variables, no higher level layer could undo that: it would be stuck with the global variables.

switch back to *main()* by returning Switching back to `main()` by returning from the fiber function has two drawbacks: it requires an internal global variable pointing to the suspended `main()` and restricts the valid use cases.

```
int main() {
    fiber_context f[] {
        ...
        // switch to 'main()' only by returning
    };
    f.resume(); // resume 'f'
    return 0;
}
```

For instance the generator pattern is impossible because the only way for a fiber to transfer execution control back to `main()` is to terminate. But this means that no way exists to transfer data (sequence of values) back and forth between a fiber and `main()`.

Switching to `main()` only by returning is impractical because it limits the applicability of fibers and requires an internal global variable pointing to `main()`.

static member function returns active fiber_context P0099R0¹⁰ introduced a static member function (`execution_context::current()`) that returned an instance of the active fiber. This allows passing the active fiber `m` (for instance representing `main()`) into the fiber `f` via lambda capture. This mechanism enables switching back and forth between the fiber and `main()`, enabling a rich set of applications (for instance generators).

```
int main() {
    int a;
    fiber_context m=fiber_context::current(); // get active fiber
    fiber_context f{[&]{
        a=0;
        int b=1;
        for(;;) {
            m=m.resume(); // switch to 'main()'
            int next=a+b;
            a=b;
            b=next;
        }
    }};
    for(int j=0; j<10; ++j) {
        f=f.resume(); // resume 'f'
        std::cout << a << " ";
    }
    return 0;
}
```

But this solution requires an internal global variable pointing to the active fiber and some kind of reference counting. Reference counting is needed because `fiber_context::current()` necessarily requires multiple instances of `fiber_context` for the active fiber. Only when the last reference goes out of scope can the fiber be destroyed and its stack deallocated.

```
fiber_context f1=fiber_context::current();
fiber_context f2=fiber_context::current();
assert(f1==f2); // f1 and f2 point to the same (active) fiber
```

Additionally a static member function returning an instance of the active fiber would violate the protection requirements of sections [encapsulating the stack](#) and [invalidation at resumption](#). For instance you could accidentally attempt to resume the active fiber by invoking `resume()`.

```
fiber_context m=fiber_context::current();
m.resume(); // tries to resume active fiber == UB
```

A static member function returning the active fiber requires a reference counted global variable and does not prevent accidentally attempting to resume the active fiber.

solution: avoiding non-const global variables and undefined behaviour

The *avoid non-const global variables* guideline has an important impact on the design of the `fiber_context` API!

synthesizing the suspended fiber The problem of global variables or the need for a static member function returning the active fiber can be avoided by **synthesizing the suspended fiber** and passing it into the resumed fiber (as parameter when the fiber is first started, or returned from `resume()`).

```
1 void foo() {
2     fiber_context f{[] (fiber_context&& m) {
3         m=std::move(m).resume(); // switch to 'foo()'
4         m=std::move(m).resume(); // switch to 'foo()'
5         ...
6     }};
7     f=std::move(f).resume(); // start 'f'
8     f=std::move(f).resume(); // resume 'f'
9     ...
10 }
```

In the pseudo-code above the fiber `f` is started by invoking its member function `resume()` at line 7. This operation suspends `foo`, empties instance `f` and synthesizes a new `fiber_context` `m` that is passed as parameter to the lambda of `f` (line 2).

Invoking `m.resume()` (line 3) suspends the lambda, empties `m` and synthesizes a `fiber_context` that is returned by `f.resume()` at line 7. The synthesized `fiber_context` is assigned to `f`. Instance `f` now represents the suspended fiber running the lambda (suspended at line 3). Control is transferred from line 3 (lambda) to line 7 (`foo()`).

Call `f.resume()` at line 8 empties `f` and suspends `foo()` again. A `fiber_context` representing the suspended `foo()` is synthesized, returned from `m.resume()` and assigned to `m` at line 3. Control is transferred back to the lambda and instance `m` represents the suspended `foo()`.

Function `foo()` is resumed at line 4 by executing `m.resume()` so that control returns at line 8 and so on ...

Class `symmetric_coroutine<>::yield_type` from N3985⁸ is **not** equivalent to the synthesized `fiber_context`.

`symmetric_coroutine<>::yield_type` does not represent the suspended context, instead it is a special representation of the same coroutine. Thus `main()` or the current thread's entry-function can **not** be represented by `yield_type` (see next section [representing `main\(\)` and thread's entry-function as fiber](#)).

Because `symmetric_coroutine<>::yield_type()` yields back to the starting point, i.e. invocation of `symmetric_coroutine<>::call_type::operator()()`, both instances (`call_type` as well as `yield_type`) must be preserved. Additionally the caller must be kept alive until the called coroutine terminates or UB happens at resumption.

This API is specified in terms of passing the suspended `fiber_context`. A higher level layer can hide that by using private variables.

representing `main()` and thread's entry-function as fiber As shown in the previous section a synthesized instance of `fiber_context` is passed into the resumed fiber.

```
int main() {
    fiber_context f{[] (fiber_context&& m) {
        m=std::move(m).resume(); // switch to 'main()'
        ...
    }};
    f=std::move(f).resume(); // resume 'f'
    ...
    return 0;
}
```

The mechanism presented in this proposal describes switching between stacks: each fiber has its own stack. The stacks of `main()` and explicitly-launched threads are not excluded; these can be used as targets too.

Thus every program can be considered to consist of fibers – some created by the OS (`main()` stack; each thread's initial stack) and some created explicitly by the code.

This is a nice feature because it allows (the stacks of) `main()` and each thread's entry-function to be represented as fibers. A `fiber_context` representing `main()` or a thread's entry-function can be handled like an explicitly created `fiber_context`: it can be passed to and returned from functions or stored in a container.

In the code snippet above the suspended `main()` is represented by instance `m` and could be stored in containers or managed just like `f` by a scheduling algorithm.

The proposed fiber API allows representing and handling `main()` and the current thread's entry-function by an instance of `fiber_context` in the same way as explicitly created fibers.

fiber returns (terminates) When a fiber returns (terminates), what should happen next? Which fiber should be resumed next? The only way to avoid internal global variables that point to `main()` is to explicitly return a non-empty `fiber_context` instance that will be resumed after the active fiber terminates.

```
1 int main(){
2     fiber_context f{[] (fiber_context&& m){
3         return std::move(m); // resume 'main()' by returning 'm'
4     }};
5     f = std::move(f).resume(); // resume 'f'
6     assert(f.empty());
7     return 0;
8 }
```

In line 5 the fiber is started by invoking `resume()` on instance `f`. `main()` is suspended and an instance of type `fiber_context` is synthesized and passed as parameter `m` to the lambda at line 2. The fiber terminates by returning `m`. Control is transferred to `main()` (returning from `f.resume()` at line 5) while fiber `f` is destroyed.

In a more advanced example another `fiber_context` is used as return value instead of the passed in synthesized fiber.

```
1 int main(){
2     fiber_context m;
3     fiber_context f1{[&] (fiber_context&& f){
4         std::cout << "f1: entered first time" << std::endl;
5         assert(!f);
6         return std::move(m); // resume (main-)fiber that has started 'f2'
7     }};
8     fiber_context f2{[&] (fiber_context&& f){
9         std::cout << "f2: entered first time" << std::endl;
10        m=std::move(f); // preserve 'f' (== suspended main())
11        return std::move(f1);
12    }};
13    std::move(f2).resume();
14    std::cout << "main: done" << std::endl;
15    return 0;
16 }
17
18 output:
19 f2: entered first time
20 f1: entered first time
21 main: done
```

At line 13 fiber `f2` is resumed and the lambda is entered at line 8. The synthesized `fiber_context f` (representing suspended `main()`) is passed as a parameter `f` and stored in `m` (captured by the lambda) at line 10. This is necessary in order to prevent destructing `f` when the lambda returns. Fiber `f2` uses `f1`, that was also captured by the lambda, as return value. Fiber `f2` terminates while fiber `f1` is resumed (entered the first time). The synthesized `fiber_context f` passed into the lambda at line 3 represents the terminated fiber `f2` (e.g. the calling fiber). Thus instance `f` is empty as the assert statement verifies at line 5. Fiber `f1` uses the captured `fiber_context m` as return value (line 6). Control is returned to `main()`, returning from `f2.resume()` at line 13.

The entry-function passed to `fiber_context`'s constructor must have signature '`fiber_context(fiber_context&&)`'. Using `fiber_context` as the return value from such a function avoids global variables.

returning synthesized `fiber_context` instance from `resume()` An instance of `fiber_context` remains empty after return from `resume()` or `resume_with()`: the synthesized fiber is returned, instead of implicitly updating the `fiber_context` instance on which `resume()` was called.

If the `fiber_context` object were implicitly updated, the fiber would change its identity because each fiber is associated with a stack. Each stack contains a chain of function calls (call stack). If this association were implicitly modified, unexpected behaviour happens.

The example below demonstrates the problem:

```

1  int main() {
2      fiber_context m, f1, f2, f3;
3      f3=fiber_context{[&](fiber_context&& f)->fiber_context{
4          f2=std::move(f);
5          for(;;){
6              std::cout << "f3 ";
7              std::move(f1).resume();
8          }
9          return {};
10     }
11     f2=fiber_context{[&](fiber_context&& f)->fiber_context{
12         f1=std::move(f);
13         for(;;){
14             std::cout << "f2 ";
15             std::move(f3).resume();
16         }
17         return {};
18     }
19     f1=fiber_context{[&](fiber_context&& f)->fiber_context{
20         m=std::move(f);
21         for(;;){
22             std::cout << "f1 ";
23             std::move(f2).resume();
24         }
25         return {};
26     }
27     std::move(f1).resume();
28     return 0;
29 }
30
31 output:
32 f1 f2 f3 f1 f3 f1 f3 f1 f3 ...

```

In this pseudo-code the `fiber_context` object is implicitly updated.

The example creates a circle of fibers: each fiber prints its name and resumes the next fiber (f1 -> f2 -> f3 -> f1 -> ...).

Fiber f1 is started at line 27. The synthesized `fiber_context` main passed to the resumed fiber is stored but not used: control flow cycles through the three fibers. The for-loop prints the name `f1` and resumes fiber f2. Inside f2's for-loop the name is printed and f3 is resumed. Fiber f3 resumes fiber f1 at line 7. Inside f1 control returns from `f2.resume()`. f1 loops, prints out the name and invokes `f2.resume()`. But this time fiber f3 instead of f2 is resumed. This is caused by the fact the instance f2 gets the synthesized `fiber_context` of f3 implicitly assigned. Remember that at line 7 fiber f3 gets suspended while f1 is resumed through `f1.resume()`.

This problem can be solved by returning the synthesized `fiber_context` from `resume()` or `resume_with()`.

```

int main() {
    fiber_context m, f1, f2, f3;
    f3=fiber_context{[&](fiber_context&& f)->fiber_context{

```

```

        f2=std::move(f);
        for(;;){
            std::cout << "f3 ";
            f2=std::move(f1).resume();
        }
        return {};
    }
};
f2=fiber_context{[&](fiber_context&& f)->fiber_context{
    f1=std::move(f);
    for(;;){
        std::cout << "f2 ";
        f1=std::move(f3).resume();
    }
    return {};
}};
f1=fiber_context{[&](fiber_context&& f)->fiber_context{
    m=std::move(f);
    for(;;){
        std::cout << "f1 ";
        f3=std::move(f2).resume();
    }
    return {};
}};
std::move(f1).resume();
return 0;
}

```

output:

f1 f2 f3 f1 f2 f3 f1 f2 f3 ...

In the example above the synthesized `fiber_context` returned by each `resume()` call is specifically move-assigned to a `fiber_context` instance other than the one on which `resume()` was called, to properly track the three fibers. (Of course this particular example depends on static knowledge of the overall control flow. But the API does not, in general, require that.)

The synthesized `fiber_context` must be returned from `resume()` and `resume_with()` in order to prevent changing the identity of the fiber.

If the overall control flow isn't known, member function `resume_with()` (see section [inject function into suspended fiber](#)) can be used to assign the synthesized `fiber_context` to the correct `fiber_context` instance (held by the caller).

```

class filament{
private:
    fiber_context      f_;

public:
    ...
    void resume_next( filament& fila){
        std::move(fila.f_).resume_with([this](fiber_context&& f)->fiber_context{
            f_=std::move(f);
            return {};
        })
    }
};

```

Picture a higher-level framework in which every fiber can find its associated `filament` instance, as well as others. Every context switch must be mediated by passing *the target* `filament` instance to *the running fiber's* `resume_next()`.

Running fiber A has an associated `filament` instance `filamentA`, whose `fiber_context` `filament::f_` is empty – because fiber A is running.

Desiring to switch to suspended fiber B (with associated filament `filamentB`), running fiber A calls `filamentA.resume_next(filamentB)`.

`resume_next()` calls `filamentB.f_.resume_with(<lambda>)`. This empties `filamentB.f_` – because fiber B is now running.

The lambda binds `&filamentA` as `this`. Running on fiber B, it receives a `fiber_context` instance representing the newly-suspended fiber A as its parameter `f`. It moves that `fiber_context` instance to `filamentA.f_`.

The lambda then returns a default-constructed (therefore empty) `fiber_context` instance. That empty instance is returned by the previously-suspended `resume_with()` call in `filamentB.resume_next()` – which is fine because `resume_next()` drops it on the floor anyway.

Thus, the running fiber’s associated `filament::f_` is always empty, whereas the `filament` associated with each suspended fiber is continually updated with the `fiber_context` instance representing that fiber.*

It is not necessary to know the overall control flow. It is sufficient to pass a reference/pointer of the *caller* (fiber that gets suspended) to the resumed fiber that move-assigns the synthesized `fiber_context` to *caller* (updating the instance).

inject function into suspended fiber

Sometimes it is useful to inject a new function (for instance, to throw an exception or assign the synthesized fiber to the caller as described in [returning synthesized fiber_context instance from resume\(\)](#)) into a suspended fiber. For this purpose `resume_with()` may be called, passing the function `fn()` to execute.

```
1 fiber_context f([](fiber_context&& caller){
2     // ...
3     std::move(caller).resume();
4     // ...
5 });
6
7 fiber_context fn(fiber_context&&);
8
9 f = std::move(f).resume();
10 // ...
11 std::move(f).resume_with(fn);
```

The `resume_with()` call at line 11 injects function `fn()` into fiber `f` as if the `resume()` call at line 3 had directly called `fn()`.

Like an entry-function passed to `fiber_context`, `fn()` must accept `std::fiber_context&&` and return `fiber_context`. The `fiber_context` instance returned by `fn()` will, in turn, be returned to `f`’s lambda by the `resume()` at line 3.

In the example below, suppose that code running on the program’s main fiber calls `resume()` (line 12), thereby entering the first lambda. This is the point at which `m` is synthesized and passed into the lambda at line 2.

Suppose further that after doing some work (line 4), the lambda calls `m.resume()`, thereby switching back to the main fiber. The lambda remains suspended in the call to `m.resume()` at line 5.

At line 18 the main fiber calls `f.resume_with()` where the passed lambda accepts `fiber_context &&`. That new lambda is called on the fiber of the suspended lambda. It is as if the `m.resume()` call at line 8 directly called the second lambda.

The function passed to `resume_with()` has almost the same range of possibilities as any function called on the fiber represented by `f`. Its special invocation matters when control leaves it in either of two ways:

1. If it throws an exception, that exception unwinds all previous stack entries in that fiber (such as the first lambda’s) as well, back to a matching `catch` clause.[†]
2. If the function returns, the returned `fiber_context` instance is returned by the suspended `resume()` or `resume_with()` call.

* *Boost.Fiber*⁴⁴ uses this pattern for resuming user-land threads.

[†] As stated in [exceptions](#), if there is no matching `catch` clause in that fiber, `std::terminate()` is called.

```

1  int data = 0;
2  fiber_context f{[&data](fiber_context&& m){
3      std::cout << "f1: entered first time: " << data << std::endl;
4      data+=1;
5      m=std::move(m).resume();
6      std::cout << "f1: entered second time: " << data << std::endl;
7      data+=1;
8      m=std::move(m).resume();
9      std::cout << "f1: entered third time: " << data << std::endl;
10     return std::move(m);
11 }};
12 f=std::move(f).resume();
13 std::cout << "f1: returned first time: " << data << std::endl;
14 data+=1;
15 f=std::move(f).resume();
16 std::cout << "f1: returned second time: " << data << std::endl;
17 data+=1;
18 f=std::move(f).resume_with([&data](fiber_context&& m){
19     std::cout << "f2: entered: " << data << std::endl;
20     data=-1;
21     return std::move(m);
22 });
23 std::cout << "f1: returned third time" << std::endl;
24
25 output:
26     f1: entered first time: 0
27     f1: returned first time: 1
28     f1: entered second time: 2
29     f1: returned second time: 3
30     f2: entered: 4
31     f1: entered third time: -1
32     f1: returned third time

```

The `f.resume_with(<lambda>)` call at line 18 passes control to the second lambda on the fiber of the first lambda.

As usual, `resume_with()` synthesizes a `fiber_context` instance representing the calling fiber, passed into the lambda as `m`. This particular lambda returns `m` unchanged at line 21; thus that `m` instance is returned by the `resume()` call at line 8.

Finally, the first lambda returns at line 10 the `m` variable updated at line 8, switching back to the main fiber.

One case worth pointing out is when you call `resume_with()` on a `fiber_context` that has not yet been resumed for the first time:

```

1  fiber_context topfunc(fiber_context&& prev);
2  fiber_context injected(fiber_context&& prev);
3
4  fiber_context f(topfunc);
5  // topfunc() has not yet been entered
6  std::move(f).resume_with(injected);

```

In this situation, `injected()` is called with a `fiber_context` instance representing the caller of `resume_with()`. When `injected()` eventually returns that (or some other) `fiber_context` instance, the returned `fiber_context` instance is passed into `topfunc()` as its `prev` parameter.

Member function `resume_with()` allows you to inject a function into a suspended fiber.

passing data between fibers

Data can be transferred between two fibers via global pointer, a calling wrapper (like `std::bind`) or lambda capture.

```

1  int i=1;
2  std::fiber_context lambda{[&i](fiber_context&& caller){

```

```

3      std::cout << "inside lambda,i==" << i << std::endl;
4      i+=1;
5      caller=std::move(caller).resume();
6      return std::move(caller);
7  });
8  lambda=std::move(lambda).resume();
9  std::cout << "i==" << i << std::endl;
10 lambda=std::move(lambda).resume();
11
12 output:
13     inside lambda,i==1
14     i==2

```

The `resume()` call at line 8 enters the lambda and passes 1 into the new fiber. The value is incremented by one, as shown at line 4. The expression `caller.resume()` at line 5 resumes the original context (represented within the lambda by `caller`).

The call to `lambda.resume()` at line 10 resumes the lambda, returning from the `caller.resume()` call at line 5. The `fiber_context` instance `caller` emptied by the `resume()` call at line 5 is replaced with the new instance returned by that same `resume()` call.

Finally the lambda returns (the updated) `caller` at line 6, terminating its context.

Since the updated `caller` represents the fiber suspended by the call at line 10, control returns to `main()`.

However, since fiber `lambda` has now terminated, the updated `lambda` is empty. Its `operator bool()` returns `false`.

Using lambda capture is the preferred way to transfer data between two fibers; global pointers or a calling wrapper (such as `std::bind`) are alternatives.

termination

Every `fiber_context` you launch must terminate gracefully by returning from its entry-function.

When an explicitly-launched fiber's entry-function returns a non-empty `fiber_context` instance, the running fiber is terminated. Control switches to the fiber indicated by the returned `fiber_context` instance. The entry-function may return (switch to) any reachable non-empty `fiber_context` instance – it need not be the instance originally passed in, or an instance returned from the `resume()` family of methods.

Calling `resume()` means: "Please switch to the indicated fiber; I am suspending; please resume me later."

Returning a particular `fiber_context` means: "Please switch to the indicated fiber; and by the way, I am done."

Cancellation of another fiber is not explicitly supported by `fiber_context`. If it is important for consuming code to communicate to a suspended fiber the desire that it should terminate, lambda binding may be used to pass some relevant object, e.g. a `stop_token`.

It is up to the code running on the fiber in question to observe and respond to any such termination request. The fiber must be resumed *after* the request before it could possibly observe the change. Even then, the entry-function might not immediately return.

One tactic would be to request termination, then loop over `resume()` or `resume_with()` calls until the returned `fiber_context` is `empty()`. However, that information is ambiguous.

Suppose we have a `fiber_context` instance `f1` representing suspended fiber F, with an application-specific termination request mechanism. The running fiber M requests F to terminate, then calls `f1.resume()`, which in due course returns another `fiber_context` instance `f2`.

`f2` has various possible values.

- `f2` might be empty. This might mean that fiber F did in fact terminate.
- Alternatively, it might mean that fiber F, instead of terminating, resumed fiber G, which terminated by resuming fiber M.
- Or fiber F might have terminated by resuming fiber G, which might have terminated by resuming fiber M.

- In other words, if `f2` is empty, fiber M cannot know the present state of fiber F.
- `f2` might not be empty. That might mean that fiber F did not terminate before resuming fiber M. `f2` would represent fiber F.
- Or it might mean that fiber F terminated by resuming fiber G, which might have resumed fiber M. `f2` would represent fiber G.
- Or it might mean that fiber F, instead of terminating, resumed fiber G, which resumed fiber M. `f2` would (again) represent fiber G.
- In other words, if `f2` is not empty, fiber M cannot know the present state of fiber F.

The `autocancel` class introduced in [Appendix D: support code for examples](#) illustrates a possible cancellation implementation, subject to the limitations described above.

exceptions

If an uncaught exception escapes from a fiber's entry-function, `std::terminate` is called.

`fiber_context` as building block for higher-level frameworks

A low-level API enables a rich set of higher-level frameworks that provide specific syntaxes/semantics suitable for specific domains. As an example, the following frameworks are based on the low-level fiber switching API of [Boost.Context](#)⁴² (which implements the API proposed here).

[Boost.Coroutine2](#)⁴³ implements **asymmetric coroutines** `coroutine<>::push_type` and `coroutine<>::pull_type`, providing a unidirectional transfer of data. These stackful coroutines are only used in pairs. When `coroutine<>::push_type` is explicitly instantiated, `coroutine<>::pull_type` is synthesized and passed as parameter into the coroutine function. In the example below, `coroutine<>::push_type` (variable `writer`) provides the resume operation, while `coroutine<>::pull_type` (variable `in`) represents the suspend operation. Inside the lambda, `in.get()` pulls strings provided by `coroutine<>::push_type`'s output iterator support.

```
struct FinalEOL{ ~FinalEOL(){ std::cout << std::endl; } };
std::vector<std::string> words{
    "peas", "porridge", "hot", "peas",
    "porridge", "cold", "peas", "porridge",
    "in", "the", "pot", "nine",
    "days", "old" };
int num=5, width=15;
boost::coroutines2::coroutine<std::string>::push_type writer{
    [&](boost::coroutines2::coroutine<std::string>::pull_type& in){
        FinalEOL eol;
        for (;;){
            for (int i=0; i<num; ++i){
                if (!in){
                    return;
                }
                std::cout << std::setw(width) << in.get();
                in();
            }
            std::cout << std::endl;
        }
    }
};
std::copy(std::begin(words), std::end(words), std::begin(writer));
```

[Synca](#)⁵⁷ (by Grigory Demchenko) is a small, efficient library to perform asynchronous operations using source code that resembles synchronous operations. The main features are a **GO-like** syntax, support for transferring execution context explicitly between different thread pools or schedulers (portals/teleports) and asynchronous network support.

```

int fibo(int v){
    if (v<2) return v;
    int v1,v2;
    Waiter()
        .go([v,&v1]{ v1=fibo(v-1); })
        .go([v,&v2]{ v2=fibo(v-2); })
        .wait();
    return v1+v2;
}

```

The code itself looks like synchronous invocations while internally it uses asynchronous scheduling.

Boost.Fiber⁴⁴ implements **user-land threads** and combines fibers with schedulers (the scheduler algorithm is a customization point). The API is modelled after the `std::thread` API and contains objects such as `future`, `mutex`, `condition_variable`...

```

boost::fibers::unbuffered_channel<unsigned int> chan;
boost::fibers::fiber f1{[&chan]{
    chan.push(1);
    chan.push(1);
    chan.push(2);
    chan.push(3);
    chan.push(5);
    chan.push(8);
    chan.push(12);
    chan.close();
}};
boost::fibers::fiber f2{[&chan]{
    for (unsigned int value: chan) {
        std::cout << value << " ";
    }
    std::cout << std::endl;
}};
f1.join();
f2.join();

```

Facebook's **folly::fibers**⁴⁷ is an asynchronous C++ framework using **user-land threads** for parallelism. In contrast to **Boost.Fiber**, **folly::fibers** exposes the scheduler and permits integration with various event dispatching libraries.

```

folly::EventBase ev_base;
auto& fiber_manager=folly::fibers::getFiberManager(ev_base);
folly::fibers::Baton baton;
fiber_manager.addTask([&]{
    std::cout << "task 1: start" << std::endl;
    baton.wait();
    std::cout << "task 1: after baton.wait()" << std::endl;
});
fiber_manager.addTask([&]{
    std::cout << "task 2: start" << std::endl;
    baton.post();
    std::cout << "task 2: after baton.post()" << std::endl;
});
ev_base.loop();

```

folly::fibers is used in many critical applications at Facebook for instance in **mcrouter**⁴⁵ and some other Facebook services/libraries like ServiceRouter (routing framework for **Thrift**⁴⁶), Node API (graph ORM API for graph databases) ...

Bloomberg's *quantum*⁴⁸ is a full-featured and powerful C++ framework that allows users to dispatch units of work (a.k.a. tasks) as coroutines and execute them concurrently using the 'reactor' pattern. Its main features are support for streaming futures which allows faster processing of large data sets, task prioritization, fast pre-allocated memory pools and parallel `forEach` and `mapReduce` functions.

```
// Define a coroutine
int getDummyValue(Bloomberg::quantum::CoroContext<int>::Ptr ctx) {
    int value;
    ...           //do some work
    ctx->yield(); //be nice and let other coroutines run (optional cooperation)
    ...           //do more work and calculate 'value'
    return ctx->set(value);
}

// Create a dispatcher
Bloomberg::quantum::Dispatcher dispatcher;
// Dispatch a work item to do some work and return a value
int result = dispatcher.post(getDummyValue)->get();
```

quantum is used in large projects at Bloomberg.

Habanero Extreme Scale Software Research Project⁵⁰ provides a task-based parallel programming model via its *HCLib*.⁵¹ The runtime provides work-stealing, `async-finish`,^{*} `parallel-for` and `future-promise` parallel programming patterns. The library is not an exascale programming system itself, but it manages intra-node resources and schedules components within an exascale programming system.

Intel's *TBB*⁵⁸ internally uses fibers for long running jobs[†] as reported by Intel.

*userver*⁵⁹ is a modern open source asynchronous framework with a rich set of abstractions, database connectors/drivers, protocols and synchronization primitives for fast and comfortable creation of IO-bound C++ microservices, services and utilities.

Alibaba's *Photon*⁵⁵ supports a large number of services and clients, especially the image service of Alibaba's container platform, which supports various Internet services for billions of users. Also used in some ByteDance services.

Alibaba's *libeasy*⁵³ supports a large number of servers, including storage, database, etc. Not officially open-sourced, but has been published as part of some open source projects, such as Oceanbase, tair, etc.

Baidu's *bthread*⁴⁹ has 1 million+ deployed instances (not counting clients) and thousands of kinds of services.

Tencent's *libco*⁵² is a c/c++ coroutine library that is widely used in backend service of WeChat, which is the largest IM service in China, with billions of users.

*libgo*⁵⁴ is developed by Meizu, one of the top mobile phone vendors in China. Libgo is used in Kiev, Meizu's distributed service framework for its applications.

*state-threads*⁵⁶ was first developed by Netscape, then maintained by SGI and Yahoo!. It is now used in a realtime media streaming server called *SRS*, and maintained by *SRS*'s developers. *state-threads* was used in the *distributed block store for Meituan*, another top Internet company in China.

As shown in this section a low-level API can act as building block for a rich set of high-level frameworks designed for specific application domains that require different aspects of design, semantics and syntax.

^{*} `async-finish` is a variant of the fork-join model. While a task might fork a group of child tasks, the child tasks might fork even more tasks. All tasks can potentially run in parallel with each other. The model allows a parent task to selectively join a subset of child tasks.

[†] because of the requirement to support a broad range of architectures `swapcontext()` was used

interaction with STL algorithms

In the following example STL algorithm `std::generate` and fiber `g` generate a sequence of Fibonacci numbers and store them into `std::vector v`.

```
int a;
autocancel consumer, generator;
generator = autocancel{ [&a, &consumer, &generator] (std::fiber_context&& m) {
    a=0;
    int b=1;
    while (! generator.stop_requested()) {
        generator.resume(consumer);
        int next=a+b;
        a=b;
        b=next;
    }
    return std::move(m);
}};
consumer = autocancel{ [&a, &consumer, &generator] (std::fiber_context&& m) {
    std::vector<int> v(10);
    std::generate(v.begin(), v.end(), [&a, &consumer, &generator] () mutable {
        consumer.resume(generator);
        return a;
    });
    std::cout << "v: ";
    for (auto i: v) {
        std::cout << i << " ";
    }
    std::cout << "\n";
    return std::move(m);
}};
consumer.resume();
```

output: v: 0 1 1 2 3 5 8 13 21 34

(See [Appendix D: support code for examples](#) for the definition of `autocancel`.)

The proposed fiber API does not require modifications of the STL and can be used together with existing STL algorithms.

possible implementation strategies

This proposal does NOT seek to standardize any particular implementation or impose any specific calling convention!

Modern **micro-processors** are **register machines**; the content of processor registers represents the execution context of the program at a given point in time.

Operating systems maintain for each process all relevant data (execution context, other hardware registers etc.) in the process table. The operating system's **CPU scheduler** periodically suspends and resumes processes in order to share CPU time between multiple processes. When a process is suspended, its execution context (processor registers, instruction pointer, stack pointer, ...) is stored in the associated process table entry. On resumption, the CPU scheduler loads the execution context into the CPU and the process continues execution.

The CPU scheduler does a **full context switch**. Besides preserving the execution context (complete CPU state), the cache must be invalidated and the memory map modified.

A kernel-level context switch is several orders of magnitude slower than a context switch at user-level.⁶

hypothetical fiber preserving complete CPU state This strategy tries to preserve the complete CPU state, e.g. all CPU registers. This requires that the implementation identifies the concrete micro-processor type and supported processor features. For instance the x86-architecture has several flavours of extensions such as MMX, SSE1-4, AVX1-2, AVX-512.

Depending on the detected processor features, implementations of certain functionality must be switched on or off. The CPU scheduler in the operating system uses such information for context switching between processes.

A fiber implementation using this strategy requires such a detection mechanism too (equivalent to `swapper/system_32()` in the Linux kernel).

Aside from the complexity of such detection mechanisms, preserving the complete CPU state for each fiber switch is expensive.

A context switch facility that preserves the complete CPU state like an operating system is possible but impractical for user-land.

fiber switch using the calling convention For `fiber_context`, not all registers need be preserved because the context switch is effected by a visible function call. It need not be completely transparent like an operating-system context switch; it only needs to be as transparent as a call to any other function. The calling convention – the part of the ABI that specifies how a function’s arguments and return values are passed – determines which subset of micro-processor registers must be preserved by the called subroutine.

The **calling convention**⁴¹ of **SYSV ABI** for **x86_64** architecture determines that general purpose registers R12, R13, R14, R15, RBX and RBP must be preserved by the sub-routine - the first arguments are passed to functions via RDI, RSI, RDX, RCX, R8 and R9 and return values are stored in RAX, RDX.

So on that platform, the `resume()` implementation preserves the **general purpose registers** (R12-R15, RBX and RBP) specified by the calling convention. In addition, the **stack pointer** and **instruction pointer** are preserved and exchanged too – thus, from the point of view of calling code, `resume()` behaves like an ordinary function call.

In other words, `resume()` acts on the level of a simple function invocation – with the same performance characteristics (in terms of CPU cycles).

This technique is used in *Boost.Context*⁴² which acts as building block for (e.g.) *folly::fibers* and *quantum*; see section `fiber_context` as building block for higher-level frameworks.

in-place substitution at compile time During code generation, a compiler-based implementation could inject the assembler code responsible for the fiber switch directly into each function that calls `resume()`. That would save an extra indirection (JMP + PUSH/MOV of certain registers used to invoke `resume()`).

CPU state on the stack Because each fiber must preserve CPU registers at suspension and load those registers at resumption, some storage is required.

Instead of allocating extra memory for each fiber, an implementation can use the stack by simply advancing the stack pointer at suspension and pushing the CPU registers (CPU state) onto the stack owned by the suspending fiber. When the fiber is resumed, the values are popped from the stack and loaded into the appropriate registers.

This strategy works because only a running fiber creates new stack frames (moving the stack pointer). While a fiber is suspended, it is safe to keep the CPU state on its stack.

Using the stack as storage for the CPU state has the additional advantage that `fiber_context` need not itself contain the stored CPU state: it need only contain a pointer to the stack location.

Section [synthesizing the suspended fiber](#) describes how global variables are avoided by synthesizing a `fiber_context` from the active fiber (execution context) and passing this synthesized `fiber_context` (representing the now-suspended fiber) into the resumed fiber. Using the stack as storage makes this mechanism very easy to implement.* Inside `resume()` the code pushes the relevant CPU registers onto the stack, and from the resulting stack address creates a new `fiber_context`. This instance is then passed (or returned) into the resumed fiber (see [synthesizing the suspended fiber](#)).

Using the active fiber’s stack as storage for the CPU state is efficient because no additional allocations or deallocations are required.

*The implementation of *Boost.Context*⁴² utilizes this technique.

`std::uncaught_exceptions()` and `std::current_exception()`

Both `std::uncaught_exceptions()` and `std::current_exception()` must report exceptions solely on the current fiber. Reporting exceptions thrown on any other fiber would make them unreliable in practice.

A straightforward implementation could make `resume()` and `resume_with()` save and restore the data underlying `std::uncaught_exceptions()` and `std::current_exception()` as part of saving and restoring the rest of the fiber state. Since `std::uncaught_exceptions()` and `std::current_exception()` data is necessarily thread-local, the likely cost would be a TLS access on every `resume()` or `resume_with()` call.

Alternatively, `fiber_context`'s constructor could update an internal associative container whose key is the high end of the new fiber stack area. `std::uncaught_exceptions()` and `std::current_exception()` could call `upper_bound()`, passing the current stack pointer, to discover which stack is current. This would shift the cost from every context switch to `std::uncaught_exceptions()` and `std::current_exception()` calls.

The examples in [Appendix A: potential premature destruction of exception object](#), [Appendix B: throw-expression with no operand](#) and [Appendix C: `std::uncaught\_exceptions\(\)` and `std::current\_exception\(\)`](#) have been floated to illustrate problems that can arise when `std::uncaught_exceptions()` and `std::current_exception()` are not specific to the current fiber.

In those small examples, the problematic code is obvious. But the power of fibers is that a function need not know whether some function it calls (or some indirect callee thereof) will resume another fiber. It's not practical simply to forbid coders from switching fibers within a catch block.

In St. Louis in June 2024, EWG requested⁶⁰ implementation experience with fiber-specific exception state.

In Wroclaw in November 2024,⁶¹ we presented a **small patch** to the Boost.Context reference implementation. With that patch, all three exception state test programs behave correctly when built with libstdc++ on Windows and Linux. Microsoft questioned whether fiber-specific exception state is implementable in MSVC, and EWG agreed to take up this matter in Hagenberg.

On February 14, 2025, Gor Nishanov stated⁶³ that a Windows Fibers implementation of `fiber_context` would be possible, while expressing concern about potential performance.

fiber switch on architectures with register window

The implementation of fiber switch is possible – many libc implementations still provide the `ucontext`-API (`swapcontext()` and related functions)* for architectures using a register window (such as SPARC). The implementation of `swapcontext()` could be used as blueprint for a fiber implementation.

how fast is a fiber switch

A fiber switch takes 11 CPU cycles on a *x86_64-Linux* system[†] using an implementation based on the strategy described in [fiber switch using the calling convention](#) (implemented in *Boost.Context*,[‡] branch *fiber*).

interaction with accelerators

For many core devices several programming models, such as OpenACC, CUDA, OpenCL etc., have been developed targeting **host-directed** execution using an attached or integrated accelerator. The CPU executes the main program while controlling the activity of the accelerator. Accelerator devices typically provide capabilities for efficient vector processing[‡]. Usually the host-directed execution uses **computation offloading** that permits executing computationally intensive work on a separate device (accelerator).⁴

For instance CUDA devices use a **command buffer** to establish communication between host and device. The host puts commands (op-codes) into the command buffer and the device processes them **asynchronously**.⁵

It is obvious that a fiber switch does **not** interact with **host-directed device-offloading**. A fiber switch works like a function call (see [fiber switch using the calling convention](#)).

*`ucontext` was removed from POSIX standard by POSIX.1-2008

[†]Intel XEON E5 2620v4 2.2GHz

[‡]warp on CUDA devices, wavefront on AMD GPUs, 512-bit SIMD on Intel Xeon Phi

multi-threading environment

Any thread in a program may be shared by multiple fibers.

A newly-instantiated fiber is not yet associated with any thread. However, once a fiber has been resumed the first time by some thread, it must thereafter be resumed only by that same thread.

There could potentially be Undefined Behaviour if:

- a function running on a fiber references `thread_local` variables
- the compiler/runtime implementation caches a pointer to `thread_local` storage in that function's stack frame
- that fiber is suspended, and
- the suspended fiber is resumed on a different thread.

The cached TLS pointer is now pointing to storage belonging to some other thread. If the original thread terminates before the new thread, the cached TLS pointer is now dangling.

For this reason, it is forbidden to resume a fiber on any thread other than the one on which it was first resumed.

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Wording

This wording is relative to N4981.⁹

Append to §3.6 [defns.block] as indicated:

[*Note 1 to entry:* Unless stated otherwise, blocking blocks the current thread. — *end note*]

Modify §4.1.2 [intro.abstract] paragraph 7.3 as indicated:

- The input and output dynamics of interactive devices shall take place in such a fashion that prompting output is actually delivered before a program an input operation waits for input. What constitutes an interactive device is implementation-defined.

Modify §6.9.2.1 [intro.multithread.general] paragraph 1 as indicated:

A *thread of execution* (also known as a *thread*) is a single flow of control the primary execution agent ([thread.req.lockable.general]) within a program, including the initial invocation of a specific top-level function, and recursively including every function invocation subsequently executed by the thread. When the host environment first enters a program, it provides a default thread to perform the program's execution steps.

When a thread is created, it runs a default fiber ([intro.fibers]).

Insert before §6.9.3 [basic.start] and renumber existing 6.9.3 to 6.9.4:

6.9.3 Fibers and Threads

[intro.fibers]

1 A *fiber* is a single flow of control within a program, including the initial invocation of a specific top-level function, and recursively including every function invocation subsequently executed by the fiber. The execution steps of a fiber are performed by a thread.

[*Note:* “Flow of control” here refers to state necessary to program execution, for example the contents of a processor's registers including its instruction pointer, and the invocation sequence ([stacktrace.general]) of functions that have been entered but have not yet returned. — *end note*]

2 A thread is always running exactly one fiber. Member functions of `fiber_context` ([fiber.context.class]) can direct the calling thread to *suspend* the running fiber and *resume* a designated other fiber. This transition from one fiber to another is a *context switch*.

3 An *implicit fiber* is the default fiber on any thread. All other fibers are *explicit fibers*.

4 An explicit fiber is created using `fiber_context`. Instantiating `fiber_context` prepares a fiber, which can consume resources. A fiber can thus be in one of three states: prepared, running or suspended.

5 When a thread first enters a prepared fiber, that thread becomes the fiber's *owning thread*. The owning thread never changes. [*Note:* A thread is the owning thread of its default fiber. — *end note*] [*Note:* If a thread resumes a fiber owned by another thread, the behaviour is undefined. — *end note*]

Modify §14.2 [except.throw] paragraph 2 as indicated:

When an exception is thrown, control is transferred to the nearest handler with a matching type ([except.handle]); “nearest” means the handler for which the *compound-statement* or *ctor-initializer* following the `try` keyword was most recently entered by the thread of control running fiber and not yet exited.

Modify §14.2 [except.throw] paragraph 4 Note 3 as indicated:

[*Note 3:* A thrown exception does not propagate to other threads fibers unless caught, stored, and rethrown using appropriate library functions; see [propagation] and [futures]. — *end note*]

Modify §14.4 [except.handle] paragraph 6 as indicated:

If no match is found among the handlers for a try block, the search for a matching handler continues in a dynamically surrounding try block of the same **thread** **fiber**.

Modify §14.4 [except.handle] paragraph 8 as indicated:

8 The exception with the most recently activated handler **in the running fiber ([intro.fibers])** that is still active is called the *currently handled exception*.

Modify §14.6.3 [except.uncaught] paragraph 1 as indicated:

... The function `std::uncaught_exceptions` (**[uncaught.exceptions]**) returns the number of uncaught exceptions in the **current thread** **running fiber ([intro.fibers])**.

Insert new final subclause in clause 33 [thread] as indicated:

33.12 fiber_context

[fiber.context]

33.12.1 Overview

[fiber.context.overview]

1 A `fiber_context` object is either *empty* or *non-empty*. A default-constructed or moved-from `fiber_context` is empty. Otherwise, a `fiber_context` is non-empty, and represents either a prepared or a suspended fiber.

2 An explicit fiber is prepared by passing an *entry-function* to `fiber_context`'s constructor. At the first call to one of the `resume()` or `resume_with()` member functions, that entry-function is entered, and the fiber is running.

3 Every call to one of the `resume()` or `resume_with()` member functions on an accessible non-empty `fiber_context` object performs a context switch.

- suspends the running fiber, making it the *previous fiber*
- resumes the fiber represented by `*this`, which was either prepared or suspended, making it the running fiber.

In addition, returning a non-empty `fiber_context` from a fiber's entry-function:

- terminates the running fiber
- resumes the fiber represented by the returned `fiber_context`.

4 When a prepared fiber is first entered, a synthesized non-empty `fiber_context` object representing the previous fiber is passed as a parameter to its entry-function. When a suspended fiber is resumed, a synthesized `fiber_context` object representing the previous fiber is returned from the relevant `resume()` or `resume_with()` member function. [Note: The synthesized `fiber_context` object received in either of those ways might represent either an explicit fiber or an implicit fiber. —end note]

5 When a running fiber returns a `fiber_context` from its entry-function, thus resuming the designated fiber, the synthesized `fiber_context` passed into the resumed fiber is empty.

6 If a fiber's entry-function returns an empty `fiber_context` object, `std::terminate` is called. If a fiber's entry-function exits via an exception, `std::terminate` is called.

7 Regardless of the number of `fiber_context` objects in the program, exactly one of them represents each prepared or suspended fiber. No `fiber_context` object represents a running fiber.

8 A `fiber_context` object can optionally be constructed by passing an explicit `span<byte>` in which to track the fiber's invocation sequence (**[stacktrace.general]**). If at any time during the life of a fiber the data storage required to track its invocation sequence exceeds the `size()` of that `span<byte>`, the behaviour is undefined.

33.12.2 Header <fiber_context> synopsis

[fiber.context.syn]

```
namespace std {  
  
    // [fibercontext], class fiber_context  
    class fiber_context;  
  
}
```

33.12.3 Class fiber_context

[fiber.context.class]

```
namespace std {  
  
class fiber_context {  
public:  
    // [fiber.context.cons], constructors, move and assignment  
    fiber_context() noexcept = default;  
  
    template<class F>  
    explicit fiber_context(F&& entry);  
  
    template<class F, class D>  
    fiber_context(F&& entry, span<byte> stack, D&& deleter);  
  
    ~fiber_context();  
  
    fiber_context(fiber_context&& other) noexcept;  
    fiber_context& operator=(fiber_context&& other) noexcept;  
  
    // [fiber.context.mem], members  
    fiber_context resume() &&;  
    template<class Fn>  
    fiber_context resume_with(Fn&& fn) &&;  
  
    bool can_resume() noexcept;  
  
    explicit operator bool() const noexcept;  
    bool empty() const noexcept;  
  
    void swap(fiber_context& other) noexcept;  
  
    // [fiber.context.special], specialized algorithms  
    friend void swap(fiber_context& lhs, fiber_context& rhs) noexcept;  
  
private:  
    void* state = nullptr;           // exposition only  
};  
  
} // namespace std
```

33.12.3.1 Constructors, move and assignment

[fiber.context.cons]

template<class F> explicit fiber_context(F&& entry) ;

1 *Constraints:*

— `remove_cvref_t<F>` is not the same type as `fiber_context`.

2 *Mandates:*

— `is_constructible_v<decay_t<F>, F>` is `true`.

— `is_invocable_r_v<fiber_context, decay_t<F>, fiber_context&&>` is `true`.

3 Effects:

- Let `entry_copy` be an object of type `decay_t<F>` direct-non-list-initialized with `std::forward<F>(entry)`.
- Initializes state to prepare a fiber that will, when first resumed, enter `entry_copy`. [*Note: `entry_copy` is not a member of `fiber_context` because it is destroyed on fiber termination, not when a `fiber_context` object is destroyed. Storage for `entry_copy` is associated with state. — end note*]
- Any necessary resources are created. [*Note: This includes storage for the new fiber's invocation sequence. — end note*]
- The prepared fiber has no owning thread.

4 Postconditions: `empty()` is `false`.

5 Throws:

- `bad_alloc` if unable to allocate storage while preparing the new fiber.
- `system_error` if unable to prepare the new fiber for any other reason.
- Any exception from initialization of `entry_copy`.

6 Error conditions: `resource_unavailable_try_again` – the system lacked the necessary resources to prepare another fiber.

template<class F, class D> fiber_context(F&& entry, span<byte> stack, D&& deleter) ;

1 Mandates:

- `is_constructible_v<decay_t<F>, F>` is `true`.
- `is_constructible_v<decay_t<D>, D>` is `true`.
- `is_invocable_r_v<fiber_context, decay_t<F>, fiber_context&&>` is `true`.
- `is_invocable_v<decay_t<D>, span<byte>>` is `true`.

2 Preconditions:

- `decay_t<D>` meets the *Cpp17MoveConstructible* requirements.
- `invoke(deleter, stack)` does not throw an exception.

3 Effects:

- Let `entry_copy` be an object of type `decay_t<F>` direct-non-list-initialized with `std::forward<F>(entry)`.
- Let `stack_copy` be a copy of `stack`. [*Note: It might be advantageous to obtain from the host environment a memory block with a read-only guard page to trap stack overflow. — end note*]
- Let `deleter_copy` be an object of type `decay_t<D>` direct-non-list-initialized with `std::forward<F>(deleter)`.
- Initializes state to prepare a fiber that will, when first resumed, enter `entry_copy`. [*Note: `entry_copy`, `stack_copy` and `deleter_copy` are not members of `fiber_context` because they are destroyed on fiber termination, not when a `fiber_context` object is destroyed. Storage for `entry_copy`, `stack_copy` and `deleter_copy` is associated with state. — end note*]
- Any necessary resources are created.
- The prepared fiber has no owning thread.

4 Postconditions: `empty()` is `false`.

5 Throws:

- `invalid_argument` if `stack.data()` fails to meet implementation-defined alignment requirements.
- `length_error` if `stack.size()` is less than the implementation-defined minimum length.
- `system_error` if unable to prepare the new fiber.
- Any exception from initialization of `entry_copy`.

— Any exception from initialization of `deleter_copy`.

6 *Error conditions*: `resource_unavailable_try_again` – the system lacked the necessary resources to prepare another fiber.

`fiber_context(fiber_context&& other) noexcept ;`

1 *Effects*: Initializes state with `exchange(other.state, nullptr)`.

`~fiber_context() ;`

1 *Effects*: If `empty()` is `false`, `terminate` is invoked ([[except.terminate](#)]).

`fiber_context& operator=(fiber_context&& other) noexcept ;`

1 *Effects*:

— If `empty()` is `false`, `terminate` is invoked ([[except.terminate](#)]).

— Equivalent to: `this->state = exchange(other.state, nullptr)`.

2 *Returns*: `*this`

33.12.3.2 Members

[[fiber.context.mem](#)]

`template<class Fn> fiber_context resume_with(Fn&& fn) && ;`

The operation of `resume_with()` involves at least two and possibly three fibers. Within [[fiber.context.mem](#)], for exposition only:

- Entering `resume_with()` performs a context switch.
- The *calling fiber* is the fiber calling `resume_with()`.
- The *target fiber* is the fiber represented by `state`.
- `resume_with()` synthesizes a `fiber_context` object representing the calling fiber. Let `caller` be that synthesized `fiber_context` object.
- Because `resume_with()` suspends the calling fiber, return from `resume_with()` necessarily requires some other fiber to perform a subsequent context switch back to the original calling fiber. When `resume_with()` returns, that other fiber is the previous fiber. [*Note*: The previous fiber can be other than the target fiber. — *end note*]
- Let `previous` be the synthesized `fiber_context` object representing the suspended previous fiber.

At entry to `resume_with()`, the target fiber can either be in the prepared state (not yet entered) or in the suspended state (waiting to return from `resume_with()`).

— If the running fiber is suspended, that implies that at some earlier time, it called `other.resume_with()`, where `other` was some non-empty `fiber_context` object. In that case, let exposition-only *internal-resume(before)*, where `before` is a `fiber_context` object, denote the following sequence of steps:

- Return `before` from `other.resume_with()`.

— Otherwise, let *internal-resume(before)* denote the following sequence of steps:

- Execute `invoke_r<fiber_context>(entry_copy, std::move(before))` and let `successor` be the resulting `fiber_context`, then
- destroy `entry_copy`, then
- if `stack_copy` and `deleter_copy` exist:
 - execute `invoke(deleter_copy, stack_copy)`, then

- destroy `deleter_copy`, then
- exit the running fiber, then
- reclaim implementation-provided resources, then
- direct the current thread to resume the fiber represented by `successor`, then
- execute `internal-resume(fiber_context())`.

1 *Mandates*: `is_invocable_r_v<fiber_context, decay_t<Fn>, fiber_context&&>` is `true`.

2 *Preconditions*: `can_resume()` is `true`.

3 *Effects*:

- Resets state so that `empty()` is `true`.
- Directs the current thread to suspend the calling fiber and resume the target fiber.
- Associates the calling thread as the target fiber's owning thread.
- Evaluates `invoke_r(std::forward<Fn>(fn), std::move(caller))`. Let returned be the `fiber_context` object returned by `fn`. [Note: returned can be other than `caller`. returned can be empty. —end note]
- Executes `internal-resume(returned)`.

4 *Returns*:

- If the previous fiber resumed the calling fiber by returning a `fiber_context` object representing the calling fiber, an empty `fiber_context`.
- If the previous fiber resumed the calling fiber by calling `resume_with(somefn)`, the `fiber_context` object returned by `invoke_r<fiber_context>(somefn, std::move(previous))`.

5 *Throws*:

If the previous fiber resumed the calling fiber by calling `resume_with(somefn)`:

- Any exception thrown by `invoke_r<fiber_context>(somefn, std::move(previous))`.

[Note: `resume_with()` throws nothing before suspending the calling fiber and ensuring `empty()` is `true`. —end note]

6 *Postconditions*: `empty()` is `true`.

[Note: Because `resume()` or `resume_with()` empties the object on which it is called, these member functions are rvalue-reference qualified. —end note]

`fiber_context resume() && ;`

1 *Effects*: Equivalent to:

`return resume_with(identity());`

`bool can_resume() noexcept ;`

1 *Returns*:

- `false` if `empty()` is `true`
- `true` if the fiber represented by `*this` is in the prepared state (has no owning thread)
- `true` if the calling thread is the owning thread of the fiber represented by `*this`
- `false` otherwise.

[Editorial note: `can_resume()` is intentionally not marked `const`. —end editorial note]

bool empty() const noexcept ;

1 *Effects*: Equivalent to: `return (! state);`

explicit operator bool() const noexcept ;

1 *Effects*: Equivalent to: `return (! empty());`

void swap(fiber_context& other) noexcept ;

1 *Effects*: Equivalent to: `swap(this->state, other.state).`

33.12.3.3 Specialized algorithms

[fiber.context.special]

friend void swap(fiber_context& lhs, fiber_context& rhs) noexcept ;

1 *Effects*: Equivalent to: `lhs.swap(rhs).`

Modify §19.6.1 [stacktrace.general] as indicated:

1 Subclause [stacktrace] describes components that C++ programs may use to store the stacktrace of the **current thread of execution** **running fiber** ([intro.fibers]) and query information about the stored stacktrace at runtime.

2 The *invocation sequence* of the current evaluation x_0 in the **current thread of execution** **running fiber** is a sequence $(x_0, \dots, x_n)$ of evaluations such that, for $i \geq 0$, x_i is within the function invocation x_{i+1} ([intro.execution]).

Header File

Add a new header file to Table 24 in §16.4.2.3 [headers]:

<fiber_context>

Feature-test Macro

Add a new feature-test macro to §17.3.2 [version.syn] as indicated:

#define __cpp_lib_fiber_context 202XXXL // also in <fiber_context>

Appendix A: potential premature destruction of exception object

In `[except.throw]` paragraph 4, the destruction of an exception object is specified to potentially occur when an active handler for the exception exits, not when a handler exits while the exception is still the currently handled exception. It is possible to observe, with the Boost implementation in an Itanium C++ ABI environment, cases where an exception is destroyed at a different point than specified (and, in particular, when a handler for the exception is still active in a fiber). Consider the following program.

```
struct Excp {
    Excp(const char *x) : x(x) {}
    ~Excp() { fprintf(stderr, "Destroying Excp(\"%s\").\n", x); }
    const char *const x;
};

int main(void) {
    // 0. fiberB is prepared but not yet resumed
    fiber_context fiberB{[] (fiber_context &&fiberA) {
        try {
            // 3. fiberB throws Excp("lambda")
            throw Excp("lambda");
        } catch (const Excp& exc) {
            // 4. fiberB catches Excp("lambda"), resumes default fiber
            fiberA = std::move(fiberA).resume();
            // 8. *** ANY ACCESS TO exc HERE ACCESSES A DESTROYED OBJECT ***
            fprintf(stderr, "9. Should destroy Excp(\"lambda\").\n");
            // 9. Excp("main") is destroyed instead
        }
        // 10. fiberB terminates by resuming default fiber
        return std::move(fiberA);
    }};
    try {
        // 1. default fiber throws Excp("main")
        throw Excp("main");
    } catch (const Excp&) {
        // 2. default fiber catches Excp("main"), enters fiberB
        fiberB = std::move(fiberB).resume();
        // 5. current_exception() reports Excp("lambda")
        fprintf(stderr, "6. Should destroy Excp(\"main\").\n");
        // 6. the current_exception() is destroyed
    }
    // 7. default fiber resumes fiberB to let it terminate
    fiberB = std::move(fiberB).resume();
}
```

Output:

```
6. Should destroy Excp("main").
Destroying Excp("lambda").
9. Should destroy Excp("lambda").
Destroying Excp("main").
```

Appendix B: throw-expression with no operand

Both `[expr.throw]` paragraph 3 and `current_exception()` (`[propagation]` paragraph 8) reference the “currently handled exception” (`[except.handle]` paragraph 8). Thus, the construct `throw;` is by definition equivalent to `std::rethrow_exception(std::current_exception());` (`[propagation]` paragraph 9).

The existing definition of currently handled exception:

“The exception with the most recently activated handler that is still active is called the *currently handled exception*.”

does not clearly constrain the scope to the current thread. This constraint must be inferred from `[except.throw]` paragraph 2:

“When an exception is thrown, control is transferred to the nearest handler with a matching type (`[except.handle]`); “nearest” means the handler for which the *compound-statement* or *ctor-initializer* following the `try` keyword was most recently entered by the thread of control and not yet exited.”

In any case, if “currently handled exception” means the exception with the most recently activated handler within any fiber on the current thread, we can get **the following result**.

```
1 struct Bad: public std::runtime_error {
2     Bad(): std::runtime_error("Bad") {}
3 };
4
5 struct Worse: public std::runtime_error {
6     Worse(): std::runtime_error("Worse") {}
7 };
8
9 int main(void) {
10     // 0. fiberB is prepared but not yet resumed
11     fiber_context fiberB{[] (fiber_context &&fiberA) {
12         try {
13             // 3. fiberB throws Worse
14             throw Worse();
15         } catch (const std::exception& caught) {
16             // 4. fiberB catches Worse, resumes default fiber
17             fiberA = std::move(fiberA).resume();
18         }
19         // 8. fiberB terminates by resuming default fiber
20         return std::move(fiberA);
21     }};
22     std::string thrown{ "Nothing" };
23     try {
24         try {
25             Bad myBad;
26             thrown = myBad.what();
27             // 1. default fiber throws Bad
28             throw myBad;
29         } catch (const std::exception& caught) {
30             // 2. default fiber catches Bad, enters fiberB
31             fiberB = std::move(fiberB).resume();
32             // 5. the most recently activated handler within the thread that
33             // is still active is in fiberB, and its exception is Worse
34             throw;
35         }
36     } catch (const std::exception& caught) {
37         // 6. caught is Worse
38         std::cout << "Situation went from " << thrown << " to " << caught.what()
39                 << std::endl;
40     }
41     // 7. default fiber resumes fiberB to let it terminate
42     fiberB = std::move(fiberB).resume();
43 }
```

Worse still, the exceptions in question aren't necessarily related to each other, and line 36 is more likely to read `catch (const Bad& caught)` – in which case the `throw;` on line 34 would *not* be caught.

Appendix C: `std::uncaught_exceptions()` and `std::current_exception()`

The following program illustrates the output of `std::uncaught_exceptions()` and `std::current_exception()` in cases involving fiber context switches within a destructor invoked by exception handling, and within a catch block.

```
fiber_context other_fiber;

void yield()
{
    assert(other_fiber);
    // We switch back and forth between two fibers, A and B. One is running,
    // the other is suspended. When fiber A calls yield():
    // 1. other_fiber is emptied
    // 2. the lambda is called on fiber B
    // 3. other_fiber is set to fiber A
    // 4. fiber B receives empty fiber_context, which is ignored
    // 5. fiber B runs for a while
    // 6. fiber B calls yield()
    // 7. other_fiber is emptied
    // 8. the lambda is called on fiber A
    // 9. other_fiber is set to fiber B
    // 10. fiber A receives empty fiber_context, which is ignored
    // 11. fiber A runs for a while...
    std::move(other_fiber).resume_with(
        [] (fiber_context&& prev)
        {
            other_fiber = std::move(prev);
            return fiber_context{};
        });
}

void uncaughts(std::string name)
{
    std::cout << " " << name << ": std::uncaught_exceptions() = "
               << std::uncaught_exceptions() << std::endl;
}

void current(std::string name)
{
    auto exc = std::current_exception();
    if (! exc)
    {
        std::cout << " " << name << ": std::current_exception() = nullptr" <<
std::endl;
    }
    else
    {
        try
        {
            std::rethrow_exception(exc);
        }
        catch (const std::exception& err)
        {
            std::cout << " " << name << ": std::current_exception() = " << err.what()
<< std::endl;
        }
    }
}

void hop(std::string name)
{
    std::cout << name << " suspending:" << std::endl;
```



```

std::string before{ name + " before" };
uncaughts(before);
current(before);
yield();
std::cout << name << " resuming:" << std::endl;
std::string after{ name + " after" };
uncaughts(after);
current(after);
}

struct destruct
{
    destruct(std::string name): mName(name + ": ~destruct()") {}
    ~destruct()
    {
        hop(mName);
    }
    std::string mName;
};

void testcode(std::string name)
{
    try
    {
        destruct d(name);
        std::string exname = name + " exception";
        std::cout << "throw " << exname << std::endl;
        throw std::runtime_error(exname);
    }
    catch (const std::exception& err)
    {
        std::cout << name << " caught " << err.what() << std::endl;
        hop(name + " catch block");
    }
}

fiber_context fiber(fiber_context&&)
{
    std::cout << "fiber() starting" << std::endl;
    testcode("fiber()");
    std::cout << "fiber() ending" << std::endl;
    assert(other_fiber);
    return std::move(other_fiber);
}

int main(int argc, char *argv[])
{
    std::cout << "main() starting" << std::endl;
    other_fiber = fiber_context(fiber);
    hop("main()");
    testcode("main()");
    std::cout << "main() ending" << std::endl;
    assert(! other_fiber);
    return 0;
}

```

With fiber-specific exception state, `std::uncaught_exceptions()` never exceeds 1, and `std::current_exception()` displays:

fiber() catch block after: `std::current_exception() = fiber() exception`

and:

```
main() catch block after: std::current_exception() = main() exception
```

Without fiber-specific exception state, `std::uncaught_exceptions()` displays up to 2 (one exception in `main()`, one in `fiber()`), and `std::current_exception()` displays:

```
fiber() catch block after: std::current_exception() = main() exception
```

and:

```
main() catch block after: std::current_exception() = fiber() exception
```

Appendix D: support code for examples

Destroying a non-empty `fiber_context` instance invokes Undefined Behaviour (see [termination](#)). To simplify code examples in this paper, we introduce an `autocancel` wrapper class that launches a fiber and tracks the sequence of `fiber_context` instances representing that fiber. When an `autocancel` instance is destroyed, it sets a stop flag and loops until the fiber voluntarily terminates.

```
// notify_done is an RAII class that binds a bool& reference and, when
// destroyed, sets the referenced bool true.
class notify_done
{
public:
    notify_done(bool& done):
        done_(done)
    {
        done_ = false;
    }

    notify_done(const notify_done&) = delete;
    notify_done& operator=(const notify_done&) = delete;

    ~notify_done()
    {
        done_ = true;
    }

private:
    bool& done_;
};

// autocancel is a wrapper class that launches a fiber and, when destroyed,
// implicitly requests stop on that fiber. It uses the tactic seen in the
// example 'filament' class to continually update the fiber_context
// representing the fiber of interest. (See "returning synthesized
// std::fiber_context instance from resume()")
class autocancel{
private:
    std::fiber_context f_;
    bool stop_flag_{false};
    bool done_{false};

public:
    autocancel() = default;
    template <typename Fn>
    autocancel(Fn&& entry_function)
    {
        f_ = std::fiber_context(
            [this, entry=std::forward<Fn>(entry_function)]
            (std::fiber_context&& prev)
            {
                notify_done term(done_);
                return entry(std::move(prev));
            });
    }

    autocancel(const autocancel&) = delete;
    autocancel& operator=(const autocancel&) = delete;
    autocancel(autocancel&&) = delete;
    autocancel& operator=(autocancel&&) = delete;

    ~autocancel() {
        stop_flag_ = true;
    }
};
```

```

        while (f_ && ! done_) {
            resume(*this);
        }
        assert(done_);
    }

    bool stop_requested() const noexcept {
        return stop_flag_;
    }

    // for initial entry from a plain fiber rather than an autocancel instance
    std::fiber_context resume() {
        return std::move(f_).resume();
    }

    void resume( autocancel& ac) {
        std::move(ac.f_).resume_with(
            [this](std::fiber_context&& f)->std::fiber_context
            {
                f_ = std::move(f);
                return {};
            });
    }
};

```

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- [28] [P0876R14: fibers without scheduler](#)
- [29] [P0876R15: fibers without scheduler](#)
- [30] [P0876R16: fibers without scheduler](#)
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